

Towards a Semantic Dataspace for Interoperable and Transparent Local Council Decision Data

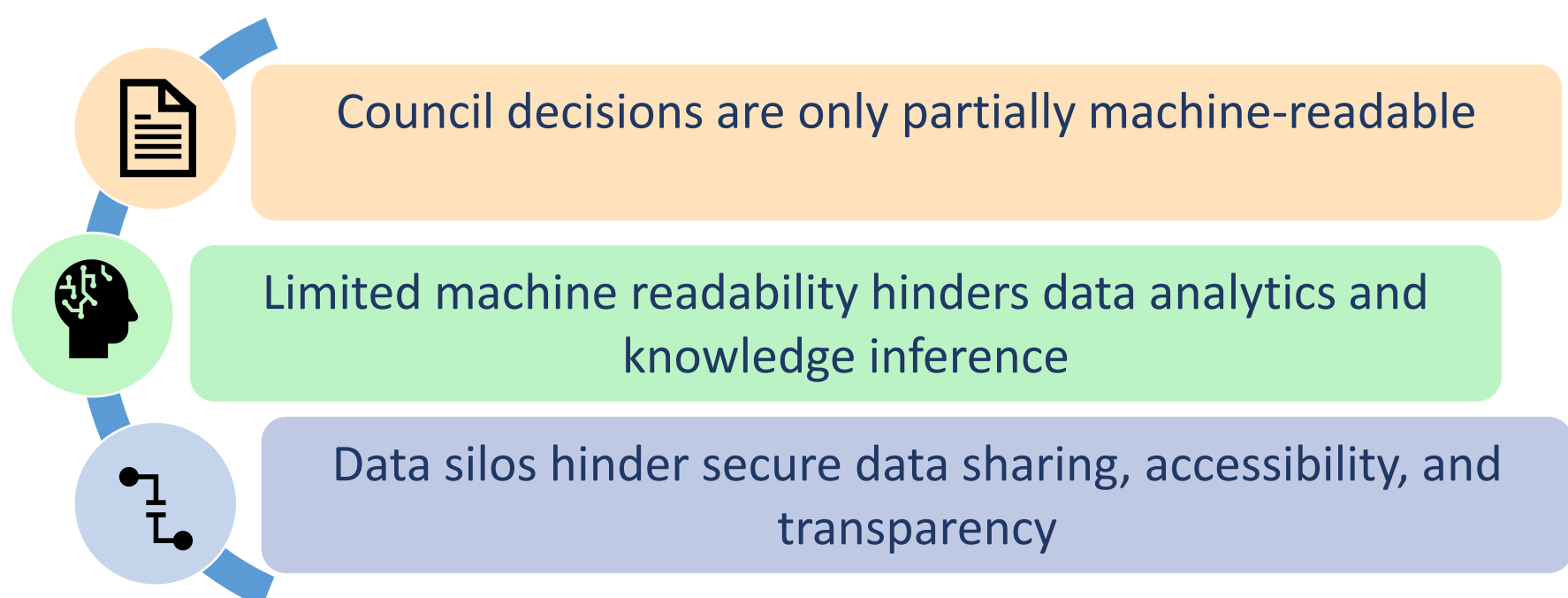
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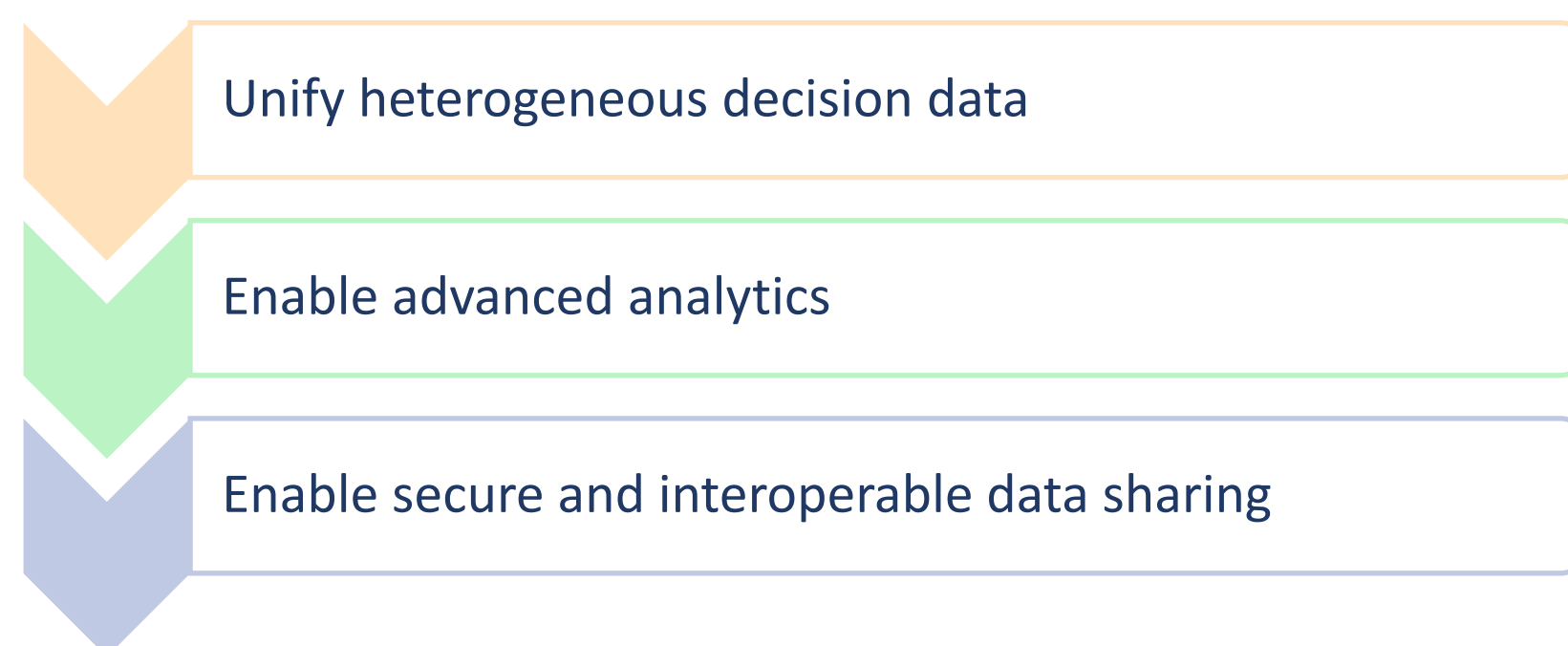
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MOTIVATION

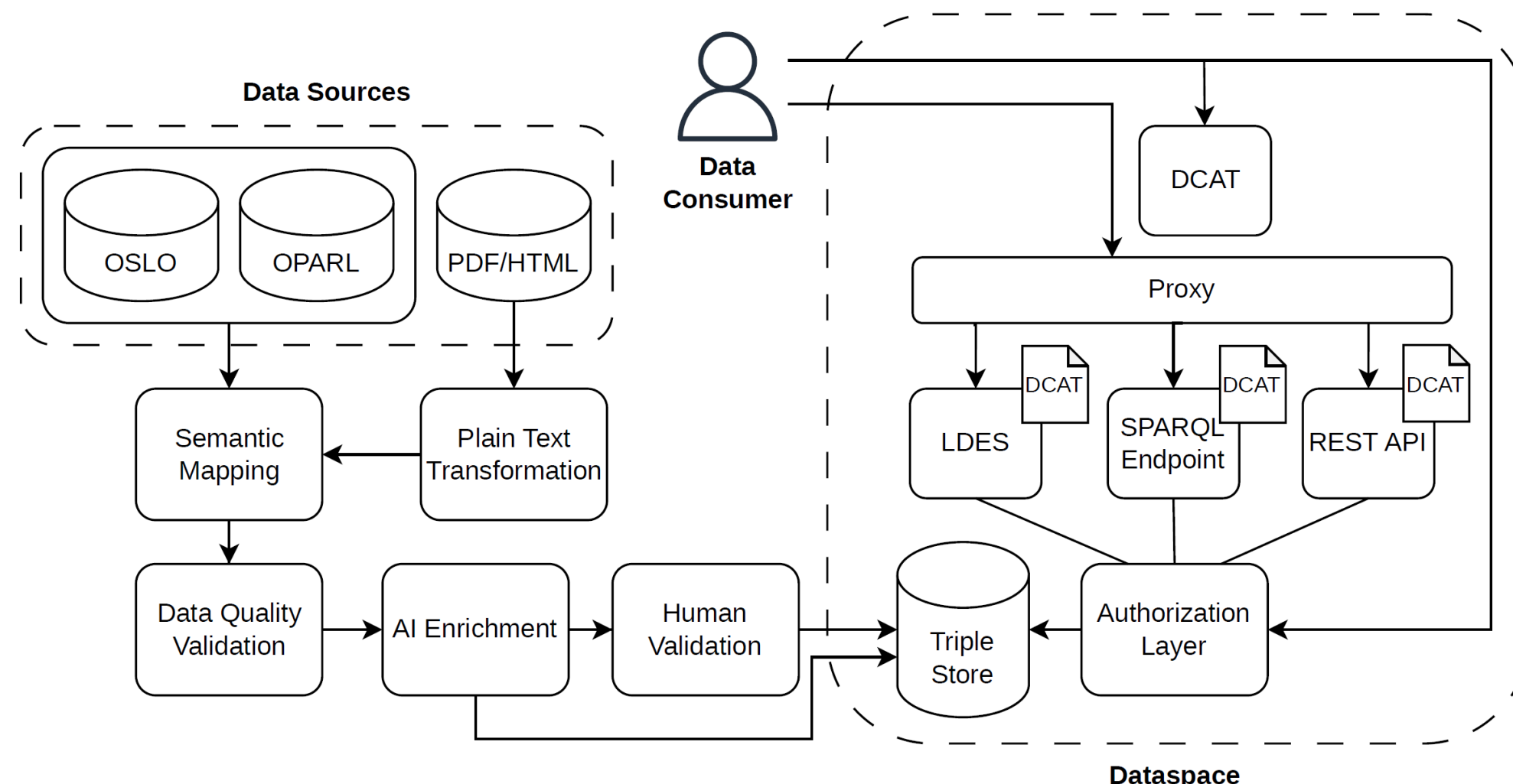


GOAL

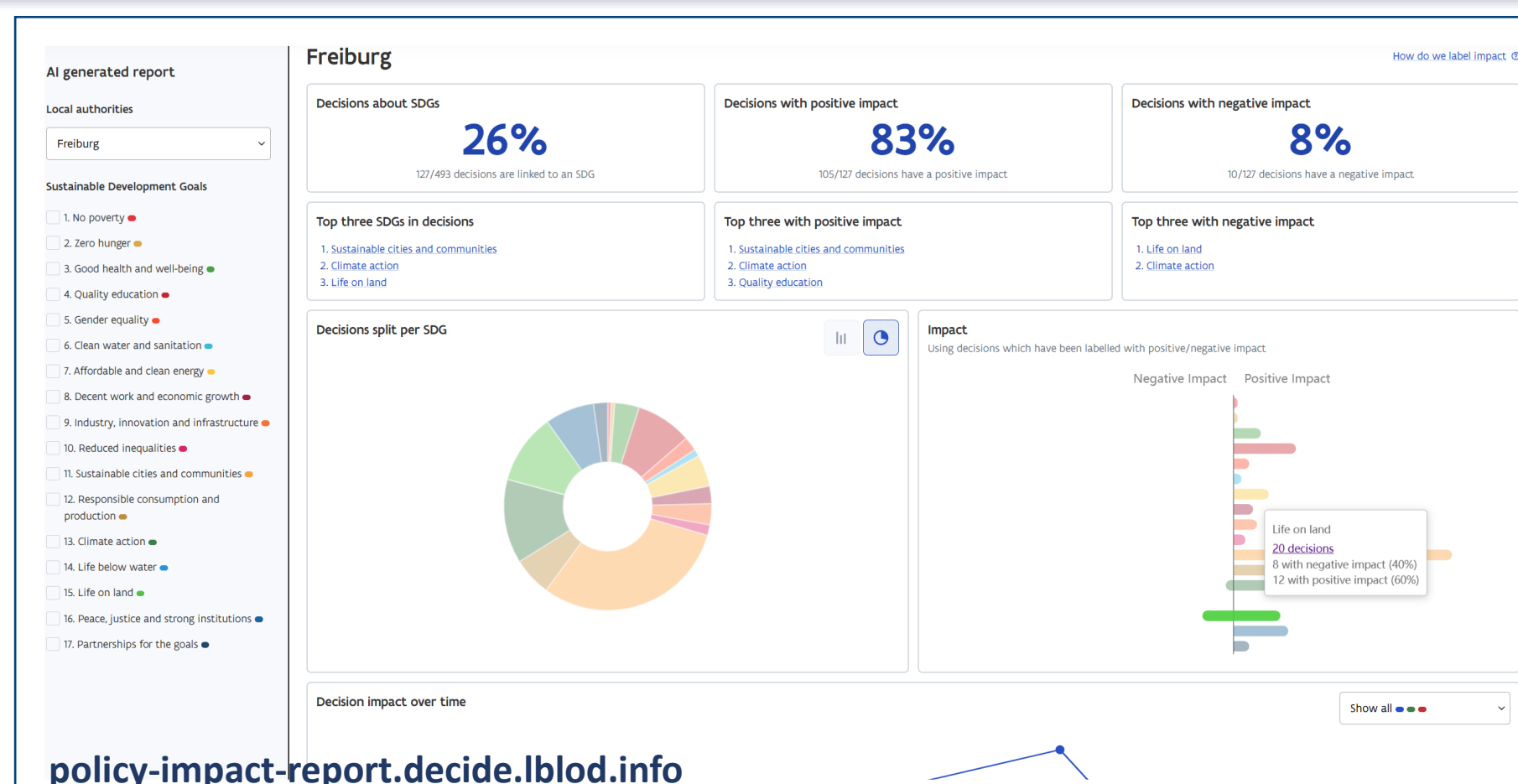


SYSTEM ARCHITECTURE

- Three data sources: Ghent (RDFa/OSLO), Freiburg (OParL API), and Bamberg (PDF/HTML).
- Map heterogeneous data to ELI and ELI-DL ontologies.
- Preprocess PDF/HTML documents (text extraction, coarse ontology mapping) and apply AI-based enrichment for entity detection.
- SHACL-based data quality validation.
- AI-enrichment:
 - Named Entity Recognition (e.g., xlm-roberta-base) and Linking using AI agents and a Model Context Protocol (MCP) Server that provides tools (e.g., querying SPARQL endpoints)
 - Linking local decisions to policy goals (e.g., the Sustainable Development Goals).
- AI-enrichments are stored directly or validated by experts.
- Discover datasets via the DCAT (Data Catalog Vocabulary) catalog.
- Securely access datasets through authenticated proxy endpoints.



USE CASES



- Goal: showcase the impact of local council decisions on policy goals.
- Each decision is linked to one or more policy goals (e.g., SDGs).
- Example: Social housing expansion → SDG 11 (Sustainable Cities and Communities, positive); industrial expansion in wetlands → SDG 15 (Life on Land, negative).
- Mapping performed in the AI-enrichment step:
 - Cold start: LLM-based classification (e.g., Mistral) without training data.
 - Supervised phase: Local classifier (e.g., DistilBERT) trained on validated annotations.
- Store the annotations in the KG and link the decision to the policy goal.
- LLM infers policy impact which are stored in the KG.
- Code: github.com/semantic-ai/codelist-labeling-service

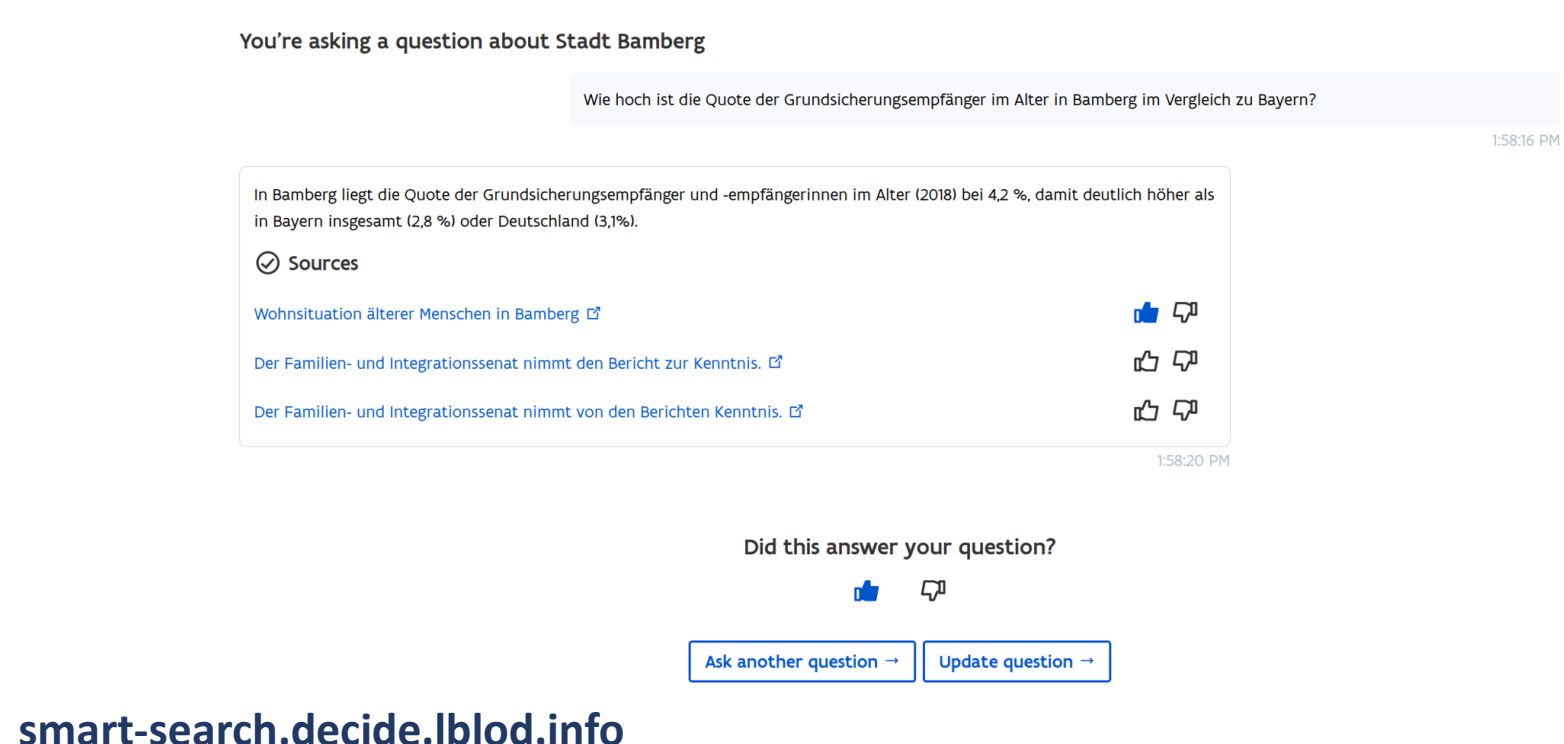
Policy Impact Report

Restricted Mobility Zones

- Goal: making local government decisions visible and accessible through geo-portals.
- Enriching decision data with location metadata.
- Performed in the AI-enrichment step:
 - Detect location entities using transformer models (e.g., XLM-RoBERTa).
 - Resolve candidate locations via geocoding (e.g., Nominatim).
 - Store geospatial entities and relations in the KG.
- Code: github.com/semantic-ai/decide-geocoding-service



DECIDE Smart Search



- Retrieval augmented generation (RAG) pipeline:
 - Embed user questions into vector representations (e.g., Ollama).
 - Retrieve semantically relevant decision documents (Decisions stored as vectors in Elasticsearch).
 - Semantically enrich retrieved documents using SPARQL query (Virtuoso triple store).
 - Generate answers with an LLM (e.g., Mistral-Nemo) using the enriched documents and the original question.
 - Output: generated answer and a list of source documents.
 - Source and answer can be validated by the user.
- Code: github.com/semantic-ai/decide-question-answering

Smart Search